

Loop Redesign of TIMP3 for Selective Binding of MMP9 Over MMP2

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Background

MMP2 and MMP9 are gelatinases with highly similar catalytic domains. MMP9 is associated with tumor invasion and MMP2 with basal basement-membrane maintenance, and broad-spectrum MMP inhibitors have not succeeded clinically, which motivates protease-selective binders [1-3]. How a binder discriminates between two nearly identical active sites is a structural question in its own right.

Tissue inhibitor of metalloproteinases 3 (TIMP3) inhibits both proteases by inserting an N-terminal ridge, formed by the AB, C, and connector loops, into the active-site cleft [4,5]. These loops were redesigned with generative models to test whether loop sequence and geometry alone can shift a scaffold's preference between homologous targets, and whether predicted interface metrics account for the preference that is measured.

Design Pipeline

Loop backbones were generated on the TIMP3 scaffold with RFDiffusion [6] (insertions of 6 to 15 residues in the AB and C loops, 4 to 10 in the EF loop; 25 backbones per target). ProteinMPNN [7] sampled sequences for each backbone, and candidates were co-folded with AlphaFold3 [8] against MMP2, MMP3, MMP9, and ADAM17.

Each variant was scored against its own cross-target mean (a self-normalized T-score), so that target-preferential variants rank above variants that score well on every target. From 549 co-fold results and 128 unique loop variants, 39 candidates were shortlisted by rank and 15 were chosen by manual curation for synthesis.

Screening and Antigen Matching

Constructs were displayed on EBY100 yeast and analyzed by flow cytometry. Display level was read from an epitope-tag stain (FITC) and target binding from APC fluorescence. The primary readout was the median APC/FITC ratio among expressing cells (Pos Med Ratio), which normalizes binding to display level. MMP9 and MMP2 signals were compared with a two-sample Welch t-test.

The three antigen suppliers do not provide equivalent molecules. Pooling trials across suppliers would compare different proteins under one target label, so selectivity was tested only on Enzo material, which supplies human catalytic-domain constructs for both targets. The two Enzo constructs span the same number of residues but are produced in different hosts (MMP2 in yeast, MMP9 in *E. coli*).

Supplier	MMP2	MMP9
Enzo	human, catalytic domain	human, catalytic domain
Sino	mouse, lacks N-catalytic half	human, full length
Masoud (in-house)	not supplied	boundaries undocumented

Conclusions

All five variants designed for MMP9 preference bound MMP9 more strongly than MMP2 on matched antigen, in every replicate, and TIMP3-WT showed the same direction. The size of the difference did not separate the variants from WT, so the measurements are compatible with the campaign producing MMP9-selective binders but do not show that the redesign, rather than the TIMP3 scaffold, is the source of the preference. The self-normalized score used to rank candidates was not tested against these measurements. The evidence is limited to two separate cultures per group measured on one day, so the fold differences and p-values are provisional. Predicted interface confidence did not explain the target-specific component of binding, which was small relative to the construct-level component, and a structural feature separating MMP9 from MMP2 binding was identified only for C 12 in the top-ranked co-fold.

Next Steps

Replication on separate days will test whether the five differences persist, estimate day-to-day variance, and include WT on every day so that variants can be compared with WT directly. Soluble expression of the TIMP3 variants is not yet available, so a whole-cell inhibition assay, compared against WT-displaying cells at matched expression, is the nearer-term proxy for activity; once soluble protein is available, surface plasmon resonance against purified, construct-matched MMP9 and MMP2 would measure dissociation constants and separate affinity from display-dependent avidity. A second design generation will score off-target binding during selection, and a sequence-based classifier (fine-tuned ESM-C) will be tested against the new measurements.

Design and Screening Workflow

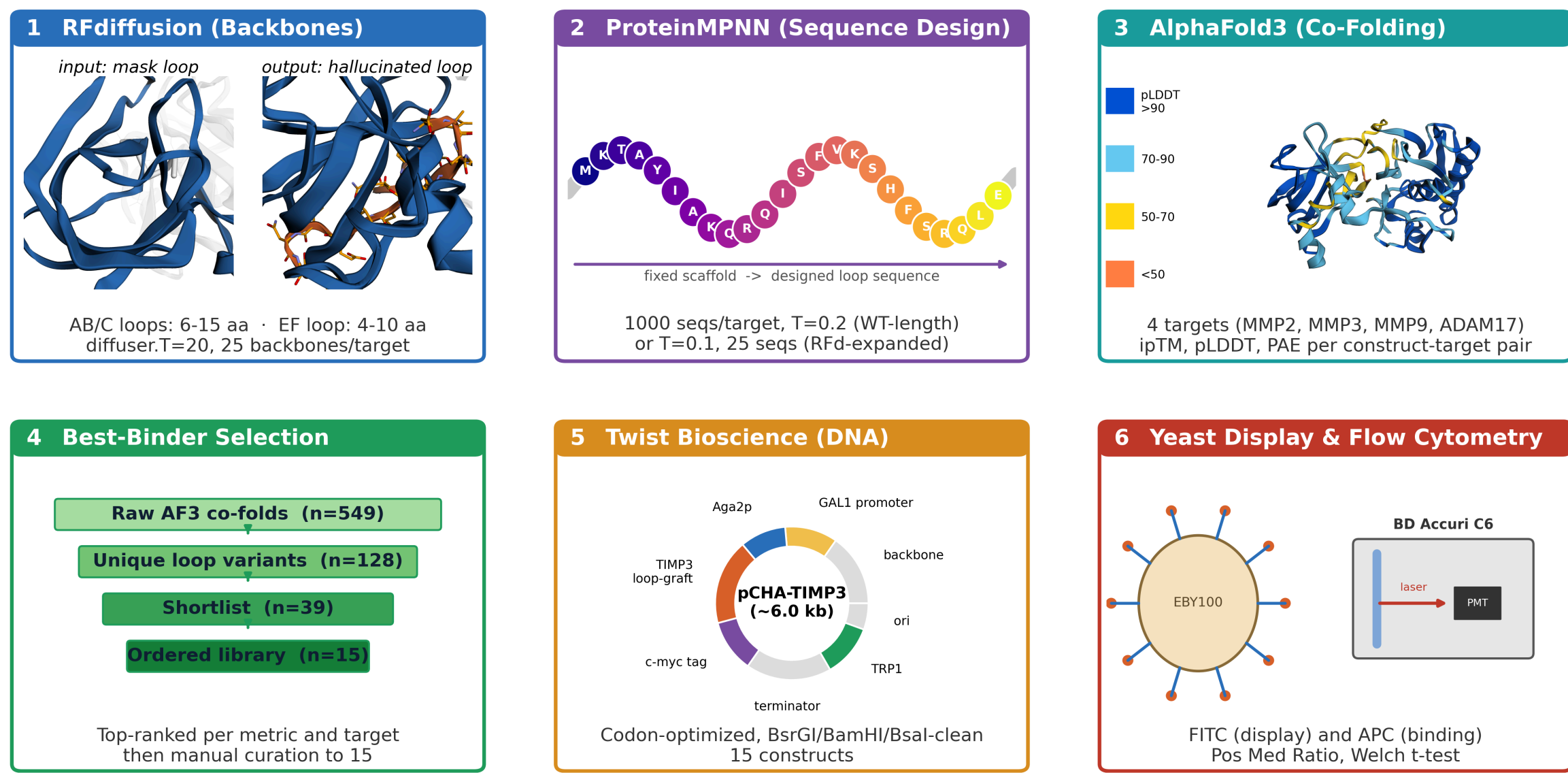


Figure 1. Design and screening workflow. Panel 1 shows the TIMP3-WT C loop masked as the RFDiffusion input (left) and the C 12 redesigned loop as output (right); panel 3 shows the C 12 co-fold with MMP9 colored by pLDDT.

Selectivity for MMP9 Over MMP2

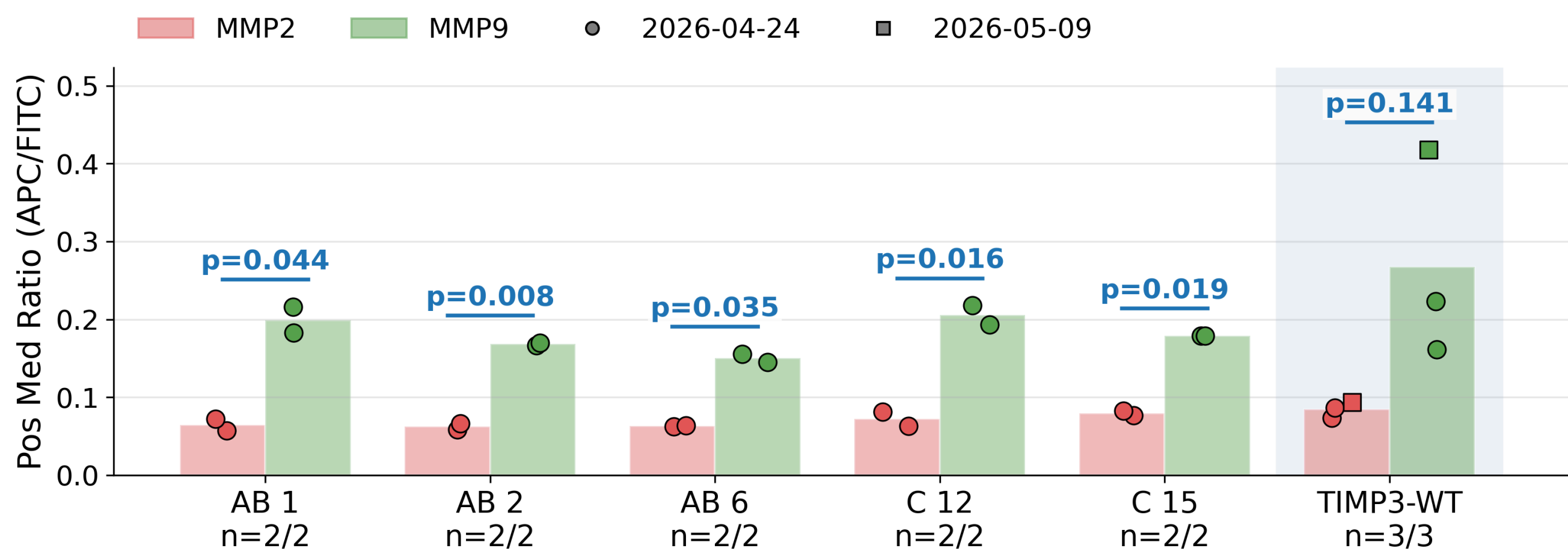


Figure 2. Pos Med Ratio for the five variants designed for MMP9 preference and TIMP3-WT on Enzo material (matched human catalytic-domain constructs). One point per separate culture; bars are means and n (MMP2/MMP9) is under each label. Brackets give nominal, uncorrected Welch p-values. Squares mark the second-day WT measurement (MMP9 0.418, against 0.161 and 0.223 on the first day), so between-day variation in one construct is comparable in size to the differences reported.

All five variants designed for MMP9 preference (AB 1, AB 2, AB 6, C 12, C 15) had higher MMP9 than MMP2 signal in every replicate, with 2.2 to 3.1-fold differences. Each group contains two separate cultures, grown in different wells and labeled with separately mixed fluorescent label preparations, all measured on one day. With five comparisons, only AB 2 falls below a Bonferroni threshold of 0.01.

TIMP3-WT showed the same direction: 2.4-fold on the same day as the variants (p = 0.16, n = 2) and 3.2-fold across all three Enzo replicates (p = 0.141, n = 3). The variants' folds were 0.93 to 1.28 times WT's fold on that day (target-by-construct interaction against WT, p = 0.32 to 1.00), and no variant had more than 7% higher MMP9 signal than WT. These measurements show that the scaffolds bind MMP9 more strongly than MMP2 in this assay; they do not separate the redesigned variants from WT.

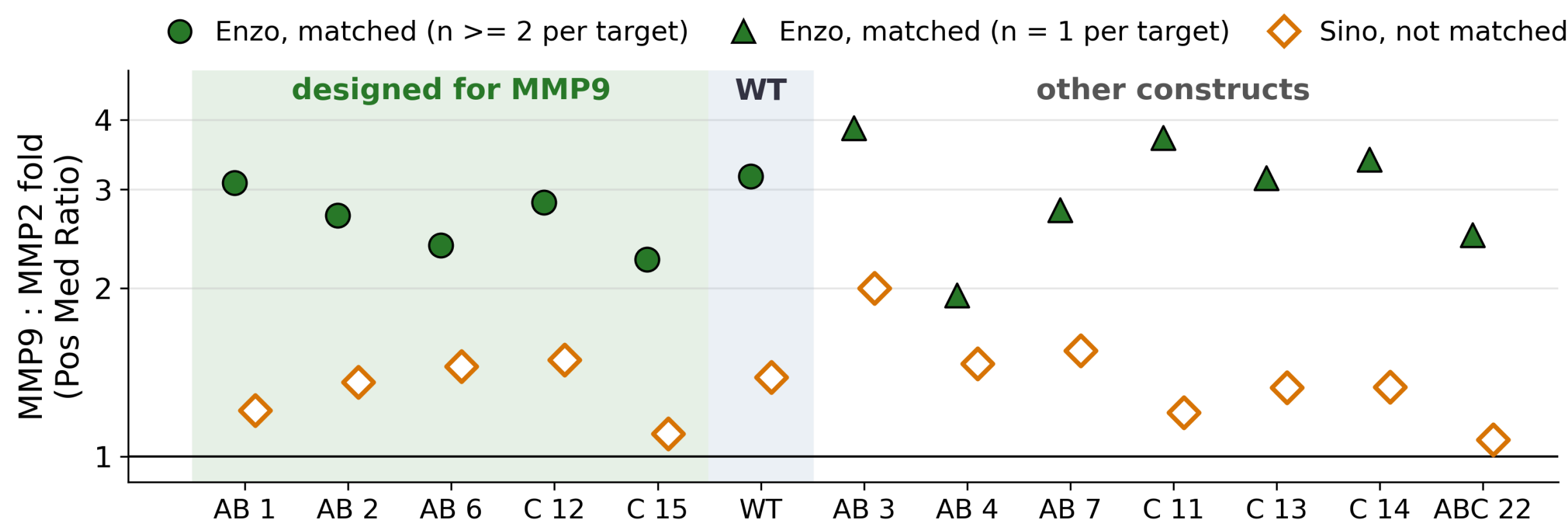


Figure 3. MMP9:MMP2 fold in Pos Med Ratio (ratio of group means) for every construct measured against both targets, by supplier. Enzo constructs are matched human catalytic domains (triangles: one trial per target, run on a different day); Sino MMP2 and MMP9 are not matched (table, left). MMP9 exceeded MMP2 for every construct at both suppliers, and the size of the difference depended on the supplier (1.1 to 2.0-fold at Sino, 1.9 to 3.9-fold at Enzo). On Enzo material the five variants (2.2 to 3.1-fold) fell within the range of the other constructs, none of which were designed for MMP9 preference.

Interface Geometry

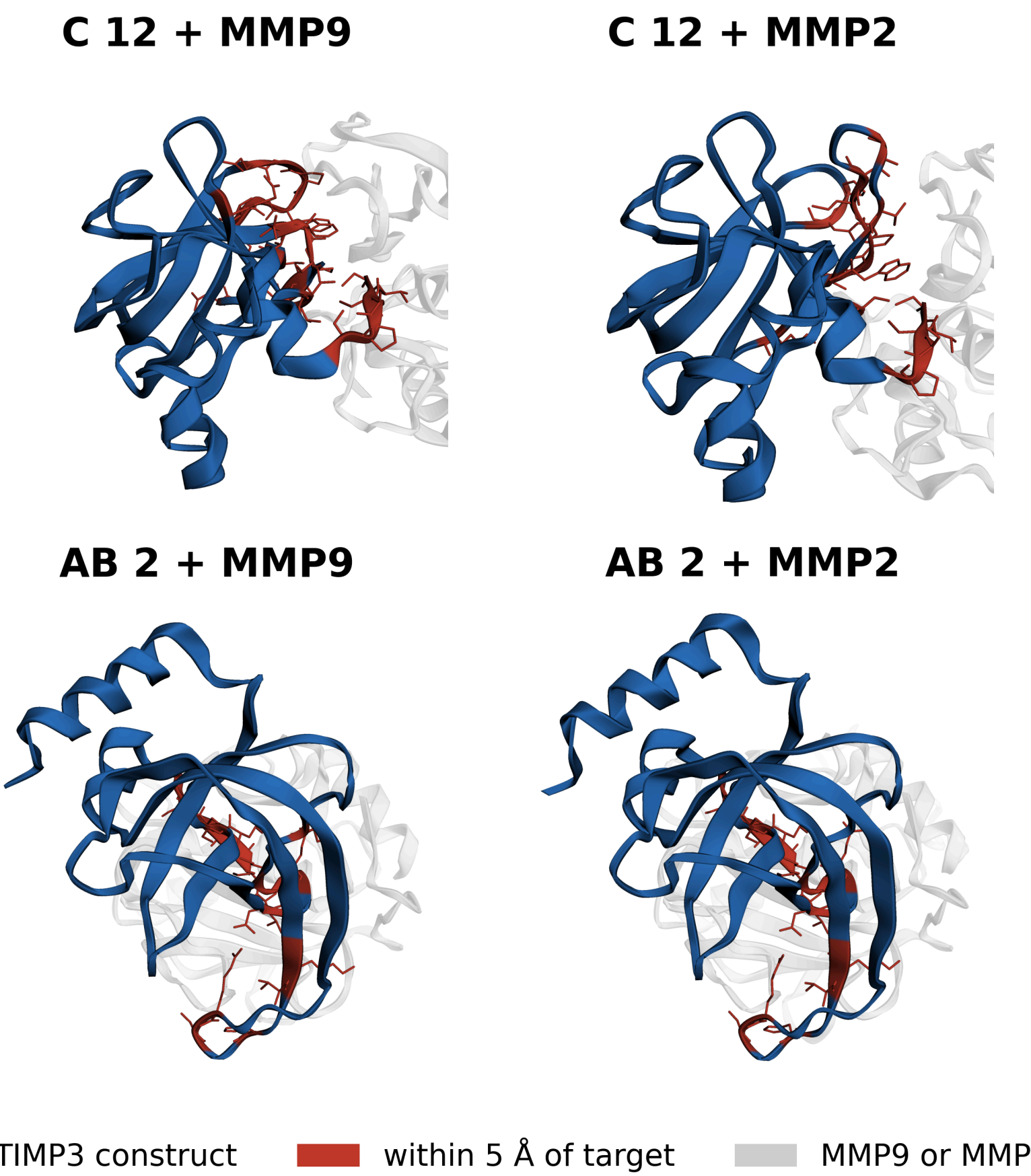


Figure 4. TIMP3-chain residues within 5 Å of the target (red) in the top-ranked AlphaFold3 co-fold of each complex. Each MMP2 co-fold is superimposed on its MMP9 co-fold by TIMP3-chain C-alpha atoms (RMSD 5.27 Å for C 12, 0.57 Å for AB 2).

Shape and charge complementarity, catalytic-loop occlusion, and buried surface area were compared between each variant's MMP9 and MMP2 co-folds. Shape complementarity differed by 0.03 or less for all five variants, and no single feature tracked selectivity across them.

C 12 was the exception in the top-ranked co-fold. Its interface contained 26 TIMP3-chain residues at MMP9 and 18 at MMP2, with a buried-surface-area difference of +180 Å², and its redesigned loop adopted a different conformation against each target. Across all five AlphaFold3 models of each complex, the C 12 interface was larger at MMP9 in four (mean difference +4 residues, range -1 to +8) and the loop conformation differed between targets in three, so the asymmetry is smaller and less consistent than the top-ranked model suggests. AB 2 contacted 19 and 18 residues in the top-ranked model (mean difference -0.4 across models) with the same loop conformation. These are static predictions and do not distinguish a structural effect from binding kinetics or avidity.

Calibration of Predicted Metrics

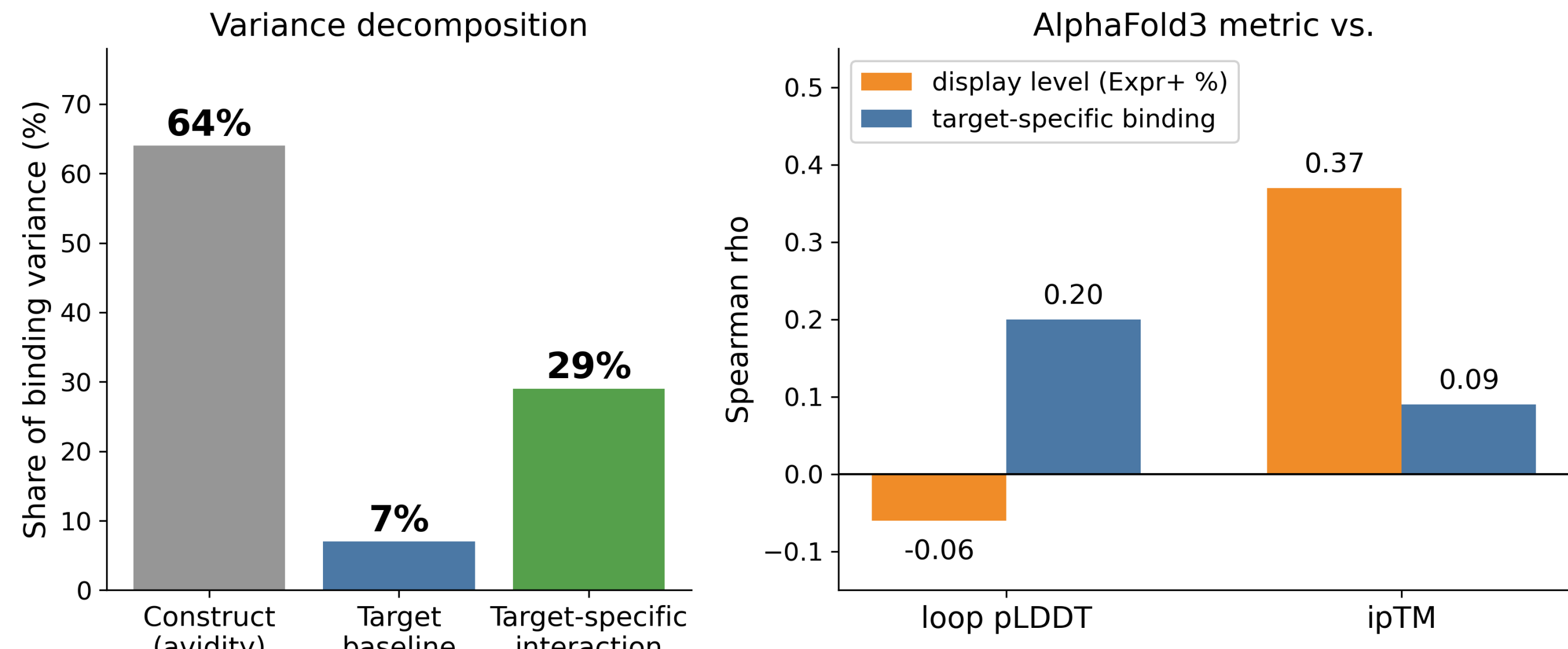


Figure 5. Left: variance decomposition of binding (12 constructs; targets MMP2, MMP9, and ADAM17). Binding is Pos Med Ratio normalized to the same-plate TIMP3-WT, pooled across suppliers. Right: Spearman correlation of AlphaFold3 metrics with display level and with the target-specific residual (n = 36 construct-target pairs; the pairs share constructs, so the p-values are approximate).

A construct-level component, which reflects general binding strength across targets, accounted for 64% of the variance in binding, the target baseline for 7%, and the target-specific interaction for 29%. After the construct component was removed, AlphaFold3 metrics did not correlate significantly with target-specific binding (ipTM rho = 0.09, p = 0.58; loop pLDDT rho = 0.20, p = 0.23). ipTM did correlate with display level (rho = 0.37, p = 0.027). This analysis pools suppliers, because only six constructs have matched MMP2 and MMP9 material, so the target components include supplier differences.

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